

Name.....

Roll No.....


ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI.
(An ISO 9001:2015 Certified College)
B. TECH DEGREE FIRST INTERNAL TEST- MARCH 2024
Eighth Semester
CST 444 & SOFT COMPUTING

Common to CSE and AI&DS

Time: 1.5 Hours

Max. Marks: 50

PART A

Answer all questions (3 x 6 = 18 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
1/	Define Artificial Neural Network, Draw its mathematical model?	3	K1	CO1&K2
2/	Compare supervised and unsupervised learning approaches in ANN	3	K2	CO1&K2
3/	Using linear separability, draw the decision boundary for logical AND and XOR?	3	K1	CO1&K2
4/	Explain the Widrow - Hoff learning rule for supervised learning in neural networks with the help of an example.	3	K2	CO2&K3
5	List out the steps in perceptron learning algorithm for single output classes.	3	K2	CO2&K3
6/	Draw the architecture of Adaline Network.	3	K2	CO2&K3

PART B

Answer any two questions (8 x 2 = 16 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
7/	Calculate the output of the neuron y for the following network using 1. binary sigmoidal activation function 2. bipolar sigmoidal activation function.	8	K2	CO1&K2

8	Draw the flowchart for the training algorithm of Hebb network.	8	K2	CO1&K2
9	Implement AND NOT function using McCulloch-Pitts neuron model. (Use binary data representation).	8	K3	CO2&K3

PART C
Answer any two questions (8 x 2 = 16 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
	Classify the two-dimensional input pattern shown in Figure using <u>perceptron</u> network. The symbol "*" indicates the data representation to be +1 and "." indicates data to be -1. The patterns are I-F. For pattern I, the target is +1, and for F, the target is -1.			
10	Figure 1-F data representation.	8	K3	CO2&K3
11	Use Adaline to train OR function with bipolar inputs and targets. Perform 2 epochs of training.	8	K3	CO2&K3
12	Draw the architecture of Back Propagation Network and write the training Algorithm.	8	K2	CO1&K2

Name.....

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ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI.
(An ISO 9001:2015 Certified College)
B. TECH DEGREE SECOND INTERNAL TEST- APRIL 2024
Eighth Semester
CST 444 & SOFT COMPUTING

Common to CSE and AI&DS

Time: 1.5 Hours

Max. Marks: 50

PART A

Answer all questions (3 x 6 = 18 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
1.	Define Fuzzy set and write basic fuzzy set operations.	3	K1	CO3&K3
2.	Consider the discrete fuzzy set defined on the universe $X = \{a, b, c, d, e\}$ as $A = \left\{ \frac{1}{a} + \frac{0.9}{b} + \frac{0.6}{c} + \frac{0.3}{d} + \frac{0}{e} \right\}$, Using Zadeh's notation, find the λ - cut sets for $\lambda = 1, 0.9, 0.6, 0+$ and 0 .	3	K3	CO3&K3
3.	Consider two fuzzy sets $A = \left\{ \frac{0.2}{0} + \frac{0.3}{1} + \frac{1}{2} + \frac{0.1}{3} + \frac{0.5}{4} \right\}$ $B = \left\{ \frac{0.1}{0} + \frac{0.25}{1} + \frac{0.9}{2} + \frac{0.7}{3} + \frac{0.3}{4} \right\}$ Find the following: (a) Algebraic sum (b) Algebraic product (c) Bounded sum.	3	K3	CO3&K3
4.	Define Genetic algorithm.	3	K1	CO4&K1
5.	Explain any three selection operators with examples.	3	K2	CO4&K1
6.	Explain the stopping conditions for the genetic algorithm.	3	K2	CO4&K1

Part B

Answer any two questions (8 x 2 = 16 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
7.	For a speed control of DC motor, the membership functions of series resistance, armature current, and speed are given as follows $SR = \left\{ \frac{0.4}{30} + \frac{0.6}{60} + \frac{1.0}{100} + \frac{0.1}{120} \right\}$ $I = \left\{ \frac{0.2}{20} + \frac{0.3}{40} + \frac{0.6}{60} + \frac{0.8}{80} + \frac{1.0}{100} + \frac{0.2}{120} \right\}$ $N = \left\{ \frac{0.35}{500} + \frac{0.67}{1000} + \frac{0.97}{1500} + \frac{0.25}{1800} \right\}$ Compute relation T for relating series resistance to Motor speed. Perform <u>Max-min</u> composition only.	8	K3	CO3&K3

8.	Using the inference approach, obtain the membership values for the triangular shapes, (L,R,T) for a triangle with angles 40° , 60° , 80° .	8	K3	CO3&K3
9.	Explain the methods used for decomposing compound linguistic rules into simple canonical rules.	8	K2	CO3&K3

PART C

Answer any two questions (8 x 2 = 16 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
10.	Describe the concept of crossover in Genetic algorithm. Explain the different methods of crossover that are possible in Genetic algorithm.	8	K2	CO4&K1
11.	Explain the different methods of encoding that are possible in Genetic algorithm.	8	K2	CO4&K1
12.	Draw the flow chart and explain the steps of Genetic Algorithm.	8	K2	CO4&K1

Reg No.:

Name:

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Eighth Semester B.Tech Degree (R, S) Examination May 2024 (2019 Scheme)

Course Code: CST444

Course Name: SOFT COMPUTING

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks.

Marks

- ✓ 1. What is the role of activation function in Artificial Neural Network (ANN)? (3)
Write and explain any two activation functions of ANN.
- ✓ 2. Compare and contrast biological neuron and artificial neuron (3)
- ✓ 3. Explain the training algorithm of Perceptron Network. (3)
- ✓ 4. Explain the architecture and the delta rule used for weight updation in Adaline network (3)
- ✓ 5. Using your own intuition, plot the Fuzzy membership function for the "Age of people"? (3)
6. What are the three basic features involved in characterizing a fuzzy membership function? Explain. (3)
- ✓ 7. Explain any 3 mutation techniques with proper examples (3)
- ✓ 8. Describe the various stopping conditions for genetic algorithm flow? (3)
- ✓ 9. Explain Dominance in Multi Objective Optimization Problem? (3)
10. What are the characteristics of Neuro Fuzzy Hybrid systems? (3)

PART B

Answer any one full question from each module, each carries 14 marks.

Module I

- 11 a) Explain linear separability with an example of AND logic implementation using Hebb network. (6)
- b) Implement AND NOT function using Mc-Culloch Pitts Neuron. Draw the architecture and threshold conditions (8)

OR

- 12 a) Using the Hebb rule, find the weights required to perform the following classifications of the given input patterns shown in figure. The "+" symbols represent the value "1" and empty space indicate "-1". Consider "1" belongs to (8)

the members of class (so has target value 1) and "O" does not belong to the members of class (so has target value -1). Perform testing of these samples also

+	+	+	+	+	+
	+		+		+
+	+	+	-	+	+
"I"			"O"		

- b) Design a two input single output biased artificial neural network for the input (6)
 X1, value 0.7 and X2, value 0.8 with weight values 0.2 and 0.3 respectively.
 The weight value associated with bias input is 0.9. Find the output Y for binary
 sigmoidal and bipolar sigmoidal activation functions.

Module II

- 13 a) State the significance of error portions δ_k , δ_j and how they are calculated in (8)
 BPN algorithm. Explain the architecture of BPN with proper labelling
 b) Implement the logic function OR with binary inputs and bipolar targets using (6)
 Perceptron network up to two epochs.

OR

- 14 a) Explain the Adaline training algorithm for single output class (6)
 b) Implement one epoch of Adaline algorithm for AND logic function with binary (8)
 inputs and bipolar outputs. Initial weights are $w_1=0.2$, $w_2=0.1$ and learning rate
 parameter $\eta=0.2$.

Module III

- 15 a) Given two fuzzy sets A and B, compute the following set operations on A (6)
 and B:
 a) Algebraic sum b) Algebraic product c) Bounded sum d) Bounded
 difference

$$A = \left\{ \frac{1}{2} + \frac{0.5}{3} + \frac{0.3}{4} + \frac{0.2}{5} \right\}$$

$$B = \left\{ \frac{0.5}{2} + \frac{0.7}{3} + \frac{0.2}{4} + \frac{0.4}{5} \right\}$$

- b) What is defuzzification? Explain different defuzzification methods with (8)
 examples.

OR

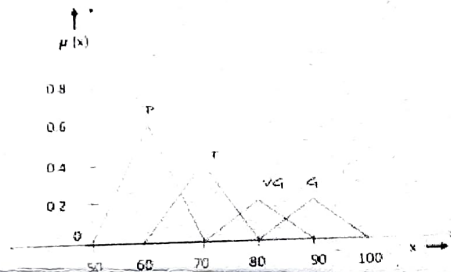
- 16 a) Consider the fuzzy relation

(5)

$$R = \begin{bmatrix} 1 & 0.8 & 0 & 0.1 & 0.2 \\ 0.8 & 1 & 0.4 & 0 & 0.9 \\ 0 & 0.4 & 1 & 0 & 0 \\ 0.1 & 0 & 0 & 1 & 0.5 \\ 0.2 & 0.9 & 0 & 0.5 & 1 \end{bmatrix}$$

Perform λ -cut operations for the values of $\lambda = 0.9, 0+$

- b) Let A be a fuzzy set that tells about a student as shown in figure below. Here, (9)
the linguistic variable P represents a Pass student, F stands for a Fair student, G represents a Good student and VG represents a Very Good student. Calculate the defuzzified value for the fuzzy set A with weighted average method and center of sums.



Module IV

- 17 a) What is the Fuzzy Inference System (FIS)? Illustrate Mamdani and Sugeno FIS (8)
with examples

- b) Explain any three selection techniques in genetic algorithm (6)

OR

- 18 a) Explain i) uniform crossover ii) Three parent crossover iii) shuffle crossover, (8)
iv) Precedence Preservative crossover of genetic algorithm with proper examples.

- b) Explain the various encoding schemes used in genetic algorithm (6)

Module V

- 19 a) Explain convex and non-convex Multi Objective Optimization Problem. How (6)
to find a non-dominated set?
- b) Explain Genetic neuro Hybrid system in detail? (8)

OR

- 20 a) What are the classifications of neuro-fuzzy hybrid systems? Discuss in detail (8)
- b) What is pareto optimality. (6)
